Fannie Mae Weekly Mortgage Applications Data: Methodology Summary

Fannie Mae's Weekly Mortgage Applications Data consist of two complementary sets of weekly indices, the Purchase Application-Level Index (PALI) and the Refinance Application-Level Index (RALI), both based on incoming mortgage application data processed through Fannie Mae's proprietary Desktop Underwriter® (DU®) application engine.[1]

Together, the PALI and RALI provide sharper insights around purchase and refinance volumes to improve transparency into the overall size of the mortgage market. The indices leverage DU loan-application data to provide market participants with current insights into the direction and scope of mortgage purchase and refinance activity. The PALI and RALI leverage DU's broad market footprint and provide a current and historical view of mortgage application activity. DU was the most widely used underwriting engine based on Home Mortgage Disclosure Act (HMDA) data from 2018 through 2024.[2] Changes over time in DU's share of total mortgage market activity may influence the interpretation of movements in the PALI and RALI. Users can reference publicly available HMDA data, which include underwriting system information, to better assess DU's share of overall mortgage market activity when evaluating trends in these indices.

Index Methodology

PALI Series Overview

Fannie Mae releases one PALI series to capture purchase application activity. In creating the index, DU applications (or casefiles) are first filtered to include only purchase applications[3] that could result in a single-family mortgage. We apply the following filters: 1) include only purchase mortgage applications; 2) include only applications that meet DU's underwriting criteria (e.g., loans that are approved and eligible for delivery to Fannie Mae, as well as loans that are approved but ineligible for delivery to Fannie Mae); 3) include only first liens; 4) exclude test cases that do not represent actual queries for purchase applications; 5) exclude duplicate applications of the same borrower with the same lender[4]; 6) include only purchase applications created in the defined week that meet the rest of the above criteria; and 7) include only purchase applications with a valid address and lender.[5]

RALI Series Overview

Fannie Mae releases three RALI series: Total RALI, Rate-Term RALI, and Cash-Out RALI. In creating the indices, DU applications (or casefiles) are first filtered to include only refinance applications[3] that could result in prepayments of existing loans and future refinances. We apply the following filters: 1) include only non-purchase mortgage applications; 2) include only applications that meet DU's underwriting criteria (e.g., loans that are approved and eligible for delivery to Fannie Mae, as well as loans that are approved but ineligible for delivery to Fannie Mae); 3) include only first liens; 4) exclude test cases that do not represent actual queries for refinance credit;

---

[1] For additional background materials, including historical time series of the data, a RALI Perspectives blog, and FAQs, visit our Data and Insights page.

[2] HMDA only began including the underwriting system information in 2018, so corresponding data prior to 2018 are not available.

[3] Because the PALI and RALI include withdrawn applications, users should follow standard methodology by considering potential changes in the "pull-through" rate, or the percentage of applications that result in origination, to arrive at the appropriate estimate of purchase and refinance and volumes.

[4] For purposes of consistency across the historical time series and comparability with lender-based survey methodologies (e.g., Mortgage Banker Association's weekly application index), we do not exclude duplicates involving the same borrower with applications at multiple lenders.

[5] For the PALI, a valid address check is performed by using Fannie Mae-specific geocode matching logic. The exclusion of applications without a valid address and lenders effectively removes prequalification loans. For RALI, there is no valid address or lender check since a refinance necessarily knows the address of the borrower requesting a refinance.

5) exclude duplicate applications of the same borrower with the same lender[4]; and 6) include only refinance applications created in the defined week that meet the rest of the above criteria.

Starting in February 2026, Fannie Mae began releasing the breakout between rate-term and cash-out refinances. This breakout is a subset of the Total RALI series, so the combined number of rate-term and cash-out applications will always equal Total RALI. Readers can therefore determine the relative shares of cash-out and rate-term refinance application volumes by dividing the respective index by the Total RALI series. A rate-term refinance is defined as a refinance that replaces an existing mortgage to change the interest rate or loan term without providing cash to the borrower. In contrast, a cash-out refinance is defined as a refinance that replaces an existing mortgage and allows the borrower to take out additional funds by leveraging home equity. The same filters outlined above are applied to both of these subsets.

Index Construction

Both the PALI and RALI are indexed before external release. The PALI and Total RALI series are indexed based on both unpaid principal balance (UPB) and loan count, with the index from the first week of 2004 (ending January 9) equaling 100, with proportional adjustments to the entire respective weekly values for both indices. Creating the indices this way allows users to track growth rates of application counts and UPB but not identify the actual underlying values. The interpretation of the index is thus relative to the value observed in the first week of January 2004. For instance, a PALI loan count value of 120 can be interpreted as having loan application counts that are 20 percent higher for the observed week relative to what was observed in the first week of January 2004. The complete weekly indexed series dating back to January 2004 for count and UPB will be provided to the market and appended with weekly releases. For the Rate-Term RALI and Cash-Out RALI, an additional indexing step is required. To allow users to view the percentage split between rate-term and cash-out refinances, we re-index these series weekly to reflect their relative weights. As a result, the first week of 2004 does not equal 100 like the Total RALI series; instead, it is adjusted to represent the split between rate-term and cash-out refinances in that week. For example, for the week ending January 2, 2026, the Rate-Term RALI UPB index is 99.7 and the Cash-Out RALI UPB index is 66.5, meaning rate-term refinances were 59.9% of all refinances. Similar to the Total RALI, the index is applied to both the UPB and count series. As small adjustments may be necessary to correct the data over time, we plan to refresh the historical weekly time series on an annual basis and notify users through our website to ensure transparency and clarity.

Timing and Release Cadence

The week of application activity starts at 12:00 a.m. Eastern Time (ET) the Saturday of that week and lasts through 11:59 p.m. ET the following Friday. We sum all applications that meet our criteria for each week and create summed values of requested loan origination amount (UPB) and summed values of application counts. We created a weekly series for data beginning with 2004, at which time the DU data are deemed reliable for PALI and RALI purposes, and apply the same filters to the historical data as to the current data so that weekly releases come from the same data-generation process as the historical series. As small adjustments may be necessary to correct the data over time, we plan to refresh the historical weekly time series on an annual basis and notify users through our website to ensure transparency and clarity; however, we undertake no obligation to do so.

The PALI and RALI are published on the Fannie Mae website each Tuesday morning, except when Monday or Tuesday is a holiday, for the week ending the previous Friday. In the event of a holiday occurring on a Monday or Tuesday, the index will be published on Wednesday for that week. Consistent with industry practice, we label the weekly index by the date the week ends (i.e., the corresponding Friday). For instance, the index values created for data captured between March 19 and March 25 would be labeled as March 25 and referred to as “the index for the week ending March 25.” The underlying data used in creation of the PALI and RALI are a direct reflection of what is observed in Fannie Mae’s DU environment, and because of this no adjustments are made for holidays or for seasonality, although our releases will note when the PALI and RALI are impacted by a holiday.